

1.Adversarial Search

(a) (3 pts) Fill in the mini-max values for each of the nodes in the tree that aren't leaf nodes

(b) (6 pts) If $L-M$ pruning were run on this tree, which branches would be cut? Mark the branches with a slash or a swirl (like a cut) and shade the leaf nodes that don't get explored.

(a) Answer (Bottom-up):

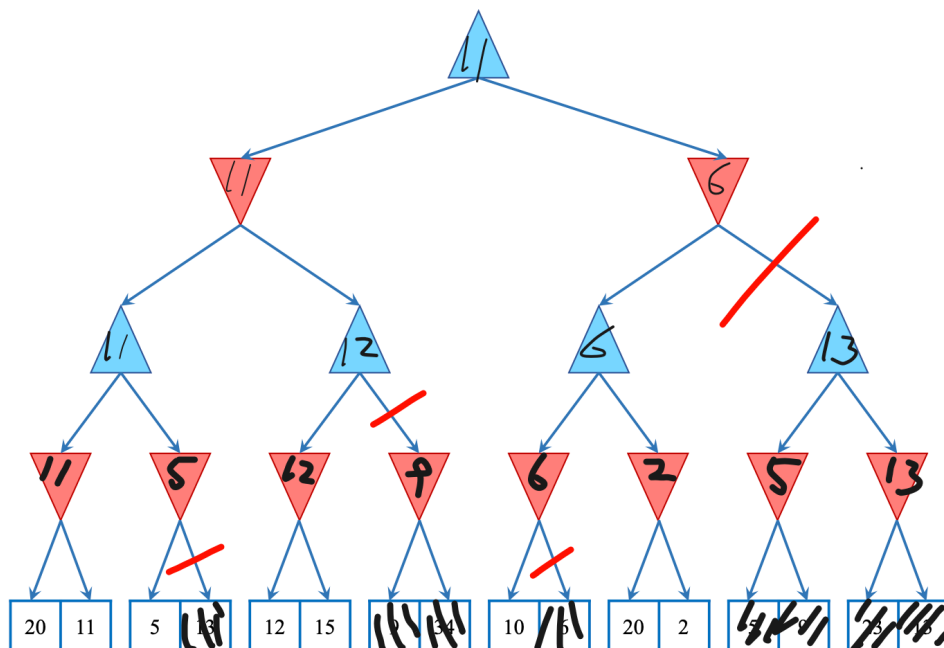
Bottom-level Min nodes (from left to right): 11, 5, 12, 9, 6, 2, 5, 13

Middle-level Max nodes (from left to right): 11, 12, 6, 13

Upper-level Min nodes (from left to right): 11, 6

Root node (Max): 11

(b)



Unexplored Leaf Nodes:

Leaves: 9, 34, 5, 9, 23, 13

2. Expectimax

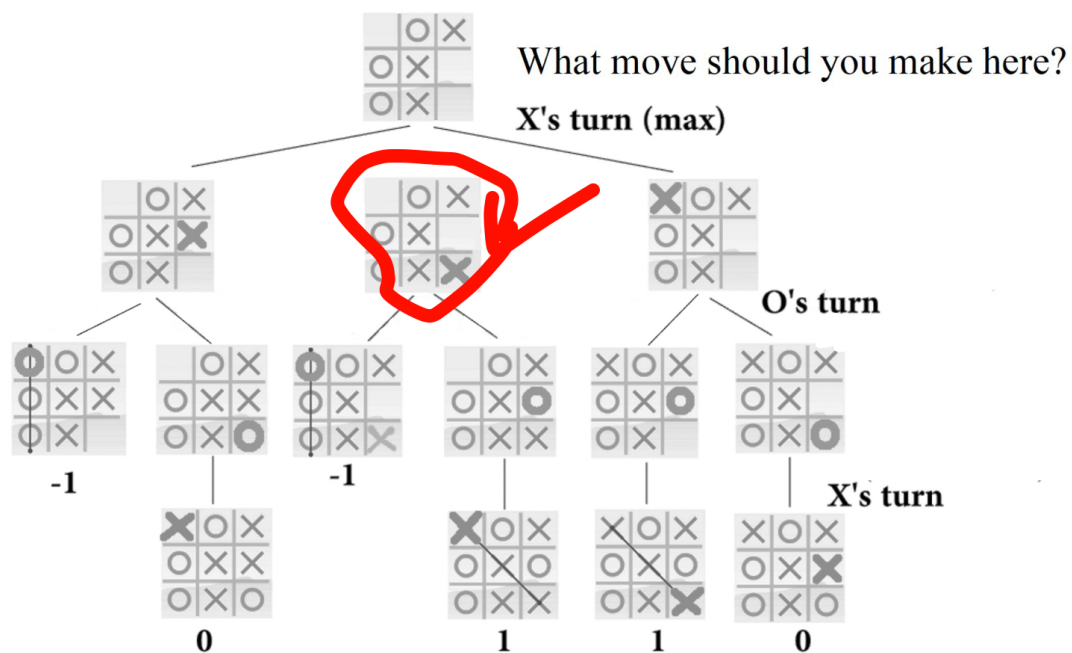
You, are playing tic tac toe as the “X” player. Y ou opponent, Olivia (“O” player) is a child, who is playing randomly. Since Olivia is very short, she is much more likely to fill “O”s into lower rows.

Specifically, you know that Olivia is three times as likely to choose a space on the middle row than the top

row, and three times again more likely to choose a space on the bottom row than the middle row. What

move should you make to maximize your chance of winning? (Circle the board position and provide justifications)

(p) :



(A) :

According to the problem description, Olivia is three times more likely to choose the middle row than the top row, and three times again more likely to choose the bottom row than the middle row. Let p be the probability of Olivia choosing a space in the Top Row.

Top Row: p , Middle Row: $3p$, Bottom Row: $9p$, Calculate the expected values of the three options.

Analysis for Move Option 1

In this scenario, placing "X" in the right of the middle row. The remaining empty spaces are Top-Left and Bottom-Right.

Total Weight: $p(\text{Top-Left}) + 9p(\text{Bottom-Right}) = 10p$

Scenario A (Olivia chooses Top-Left): Olivia completes a vertical line and wins.

Utility: -1, Probability: $\frac{p}{10p} = \frac{1}{10} = 0.1$

Scenario B (Olivia chooses Bottom-Right): "X" is blocked; the subsequent move to Top-Left leads to a draw.

Utility: 0, Probability: $\frac{9p}{10p} = \frac{9}{10} = 0.9$

$EV = (0.1 \times -1) + (0.9 \times 0) = -0.1$

Analysis for Move Option 2

In this scenario, placing "X" in the Bottom-Right position. The remaining empty spaces are Top-Left (Weight p) and Middle-Right (Weight $3p$), Total Weight: $4p$.

Scenario A (O plays Top-Left): Olivia completes a vertical line and wins.

Utility: -1, Probability: $P = \frac{p}{4p} = 0.25$

Scenario B (O plays Middle-Right): "X" blocks Olivia and wins on the next turn.

Utility: 1 Probability: $P = \frac{3p}{4p} = 0.75$

$EV = (0.25 \times -1) + (0.75 \times 0) = 0.5$

Analysis for Move Option 3

In this scenario, placing "X" in the Top-Left position. The remaining empty spaces are Middle-Right (Weight $3p$) and Bottom-Right (Weight $9p$), Total Weight: $12p$.

Scenario A (O plays Middle-Right): "X" can win on the next turn.

Utility: 1, Probability: $P = \frac{3p}{12p} = 0.25$

Scenario B (O plays Bottom-Right): Leads to a draw.

Utility: 0 Probability: $P = \frac{9p}{12p} = 0.75$

$EV = (0.25 \times 1) + (0.75 \times 0) = 0.25$

The optimal move is Option 2 (placing "X" in the Bottom-Right corner), as it yields the highest expected value 0.5.
